Samaranjan Manjhi

C/C++ Systems & Cyber Security Software Developer

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SUMMARY

C/C++ Systems and Cyber Security Engineer with **3+ years** of experience building production-grade endpoint security tools, kernel monitors, binary integrity frameworks, and Data Loss Prevention (DLP) systems for **macOS and Linux**. Deep expertise in Endpoint Security Framework (ESF), Elliptic Curve cryptographic signing, execution gating, system call hooking, Qt GUI development, and cross-platform C++ architecture. Delivered **10+ enterprise security modules** deployed globally across macOS and Linux endpoints for enterprise antivirus suites.

TECHNICAL SKILLS

Languages: C, C++, Bash Scripting, Relational SQL
Frameworks: Qt Framework, Endpoint Security Framework (ESF), Extended Berkeley Packet Filter, System Audit, Core Foundation
Security: Cyber Security Architecture, Elliptic Curve Cryptography, Sodium Cryptography Library, OpenSSL Library, Packet Filter Firewall
Tools: Git, CMake, Make, Xcode, GDB, Valgrind, KDB/KGDB, Ftrace/DTrace, Kdump, SQLite, spdlog, XML Parsers, Boost, Nano
Core CS: Data Structures, Multithreading, Inter-Process Communication, Operating System Internals, Linux, macOS

PROFESSIONAL EXPERIENCE

Systems & Security Software Engineer | **MicroWorld Software Services Pvt. Ltd. — Mumbai** *Aug 2023 – Present*

- Designed and delivered **10+ production security modules** spanning kernel monitors, DLP engines, binary integrity systems, firewall rule generators, and cloud backup agents — deployed globally across enterprise Linux and macOS systems running security products.
- Constructed high-performance cross-platform security daemons in C/C++, integrating Linux kernel modules, macOS ESF event loops, and POSIX Unix domain socket IPC to reduce system memory overhead by 25% and improve event detection throughput by 40%.
- Independently owned end-to-end design, implementation, and deployment for select modules, coordinating with QA and product teams to define requirements and validate detection accuracy prior to release.
- Regularly mentored and onboarded new engineers on secure C/C++ development, kernel-level debugging, and internal security frameworks, shortening ramp-up time on kernel and DLP module development.

KEY PROJECTS

Real-Time Virus Scanning — eBPF | **C/C++, eBPF, Kernel Tracing, AV Engine Integration** *Mar 2026 – Jun 2026*

- Built a real-time on-access antivirus scanning daemon using eBPF programs attached to kernel hooks to intercept file open, write, and execution events, streaming new and modified files into the AV scan engine for detection.
- Integrated quarantine and process-termination actions triggered on malware detection, with a configurable watchlist and root/parent-process validation to reduce false positives and redundant re-scans.
- Optimized event throughput with a lock-free audit map and dedicated scanner thread pool, sustaining real-time on-access detection under heavy file I/O without blocking user processes.

MW Guardian — Binary Integrity & Execution Gating | **C/C++, Linux Kernel, Cryptographic Verification** *Feb 2026 – Present*

- Enforced execution gating via **kernel permission events** to intercept binary execution at kernel level, synchronously validating cryptographic signatures and terminating untrusted binaries prior to execution.
- Architected a multi-key cryptographic signing pipeline supporting key rotation while retaining legacy public keys to preserve signature validity for previously signed binaries.
- Created startup self-verification routines where guardian and verifier components validate their own binary signatures prior to initiating the main event loop, eliminating bootstrap trust gaps.

Linux Ransomware Defense — Kernel Module | **C, Linux Kernel, Syscall Hooking, Behavior Tracking** *Dec 2025 – Feb 2026*

- Intercepted file rename, execution, and open system calls using kernel tracing to detect ransomware behavior via threshold-based bulk rename tracking and suffix pattern matching, terminating offending processes and logging security events.
- Enhanced driver stealth by hiding kernel module entries from the active driver list post-load and restricting module unload operations through a password-authenticated system interface.
- Provided runtime configurability at initialization time, allowing dynamic updates to max-rename thresholds, behavior detection toggles, and passphrases without requiring kernel re-compilation.

Printer Control & Data Loss Prevention (DLP) | **macOS, C/C++, ESF, Print Service Libraries, Xcode** *Oct 2025 – Jan 2026*

- Developed a macOS DLP driver utilizing ESF and print spool monitoring to inspect print files for sensitive information, cancelling unauthorized print jobs prior to printing.
- Applied application-level process whitelisting and blacklisting enforced via ESF process identity, bypassing clean content for trusted applications and blocking unauthorized applications outright.

File Integrity Monitoring (FIM) | **macOS, C/C++, File System Events, SQLite Database, OpenSSL** *Jun 2025 – Aug 2025*

- Built a file system event daemon storing cryptographic file hashes, permissions, file sizes, and timestamps in SQLite, emitting real-time security alerts upon unauthorized path modifications, renames, or deletions.
- Established baseline state snapshots on initial startup and executed diff comparisons on subsequent runs to output structured security event logs across configurable watch paths.

Content Scanner — DLP Engine & CLI | **C/C++, Regex Engine, Optical Character Recognition, Archive Libraries** *Sep 2024 – May 2025*

- Built a multi-threaded DLP scanner supporting **15+ file formats** via regular expression parsing and optical character recognition, packaged as a shared library and CLI utility.
- Developed a high-performance Unix domain socket server variant allowing clients to stream file paths and retrieve scan results without re-initializing regex engines, reducing per-scan latency by 60%.

COMPETITIVE PROGRAMMING & PROBLEM SOLVING

Solved Data Structures and Algorithms problems (Arrays, Dynamic Programming, Sliding Window, Graphs, Memory Management) across online judges: **LeetCode** | **GeeksForGeeks** | **Coding Ninjas**

EDUCATION

Bachelor of Engineering — Computer Engineering
Shree L.R. Tiwari College of Engineering, Mumbai University — CGPA: **8.56 / 10**